Network project

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Project Overview

We are pleased to present this comprehensive documentation for our network design project, developed using **Cisco Packet Tracer**. The project aims to simulate and design a modern, scalable enterprise network that adheres to the standard **Hierarchical Engineering** model to ensure **optimal performance, high reliability, and enhanced security.**The design focuses on implementing advanced **routing concepts**, **network security**, and high availability (Redundancy) through protocols such as **OSPF, HSRP, and EtherChannel** and secures access using **SSH and AAA.**

The network design includes multiple layers following industry standards:

The designed network consists of three main layers: the **Core, Distribution, and Access layers.** Routers and Multilayer Switches were utilized to implement crucial tasks such as **Inter-VLAN Routing,** in addition to efficiently segmenting the network into multiple **Virtual Local Area Networks (VLANs).**

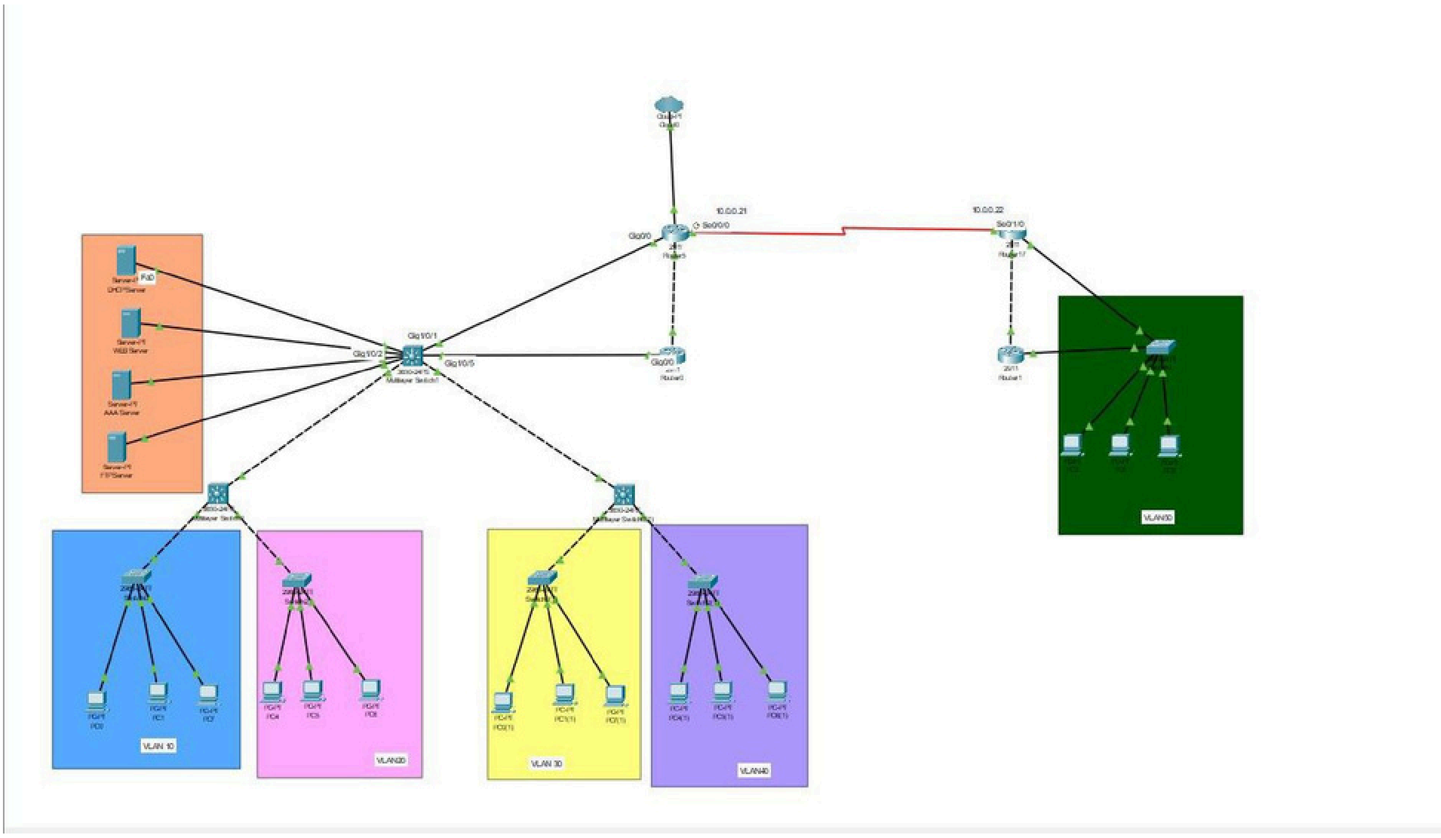
This documentation aims to explain the details of the infrastructure, the IP addressing scheme, and the key configuration processes that ensure the network operates as an integrated and effective unit.

Key technologies and configurations implemented in this project include:

The network was designed to meet the following requirements:

1. Use variable subnetting.
2. Use OSPF as the routing protocol.
3. Implement VLANs and inter-VLAN routing using ROAS.
4. Use a standalone DHCP server for network configuration.
5. Configure AAA for authentication with a local admin user.
6. Set the privilege exec mode password to the group number.
7. Use a web server with a public IP (Static NAT).
8. Configure EtherChannel for switch redundancy.
9. Use HSRP for router redundancy.
10. Backup all configurations to an FTP server.
11. Enable SSH on at least one router.

Topology Design

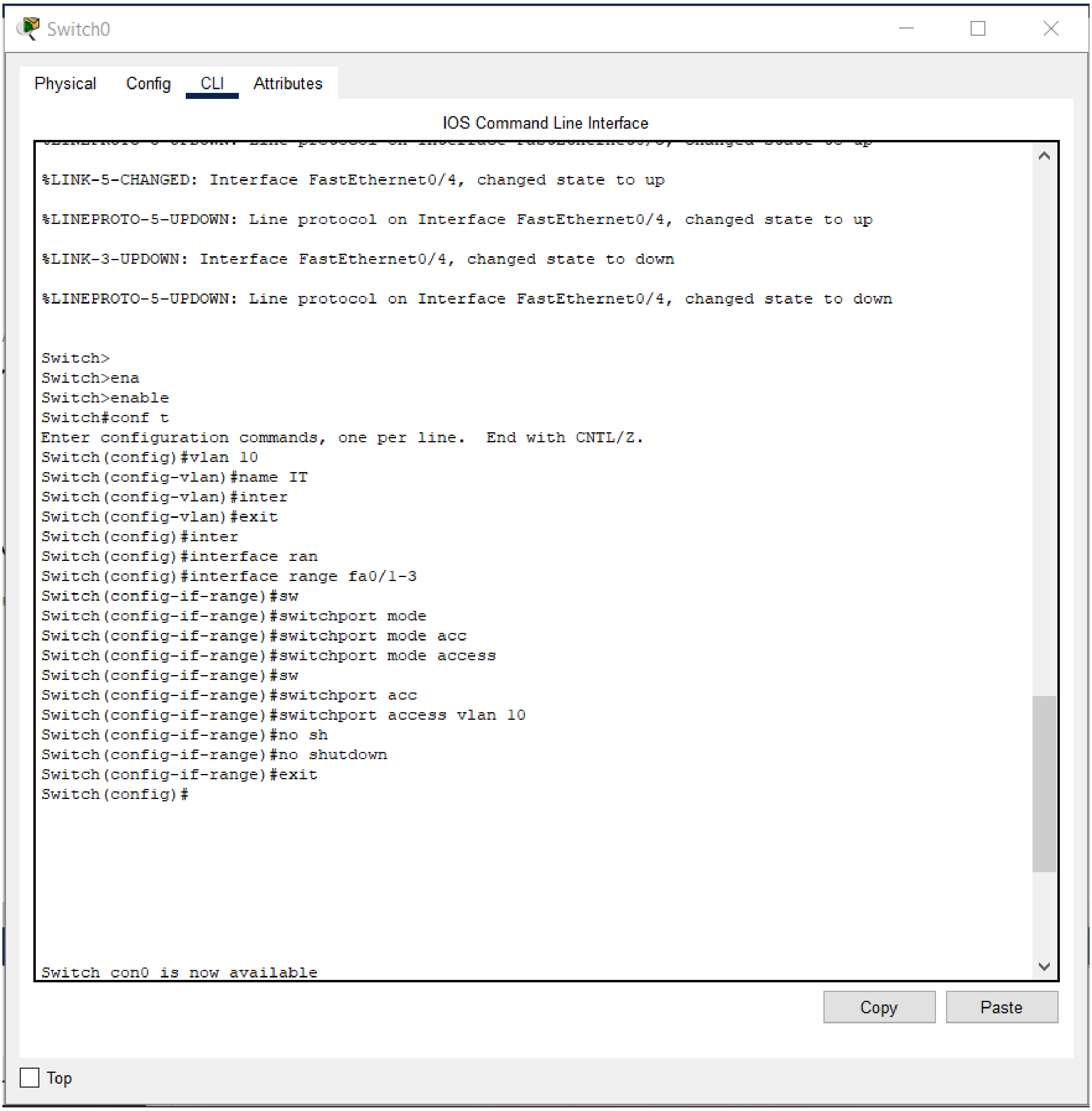


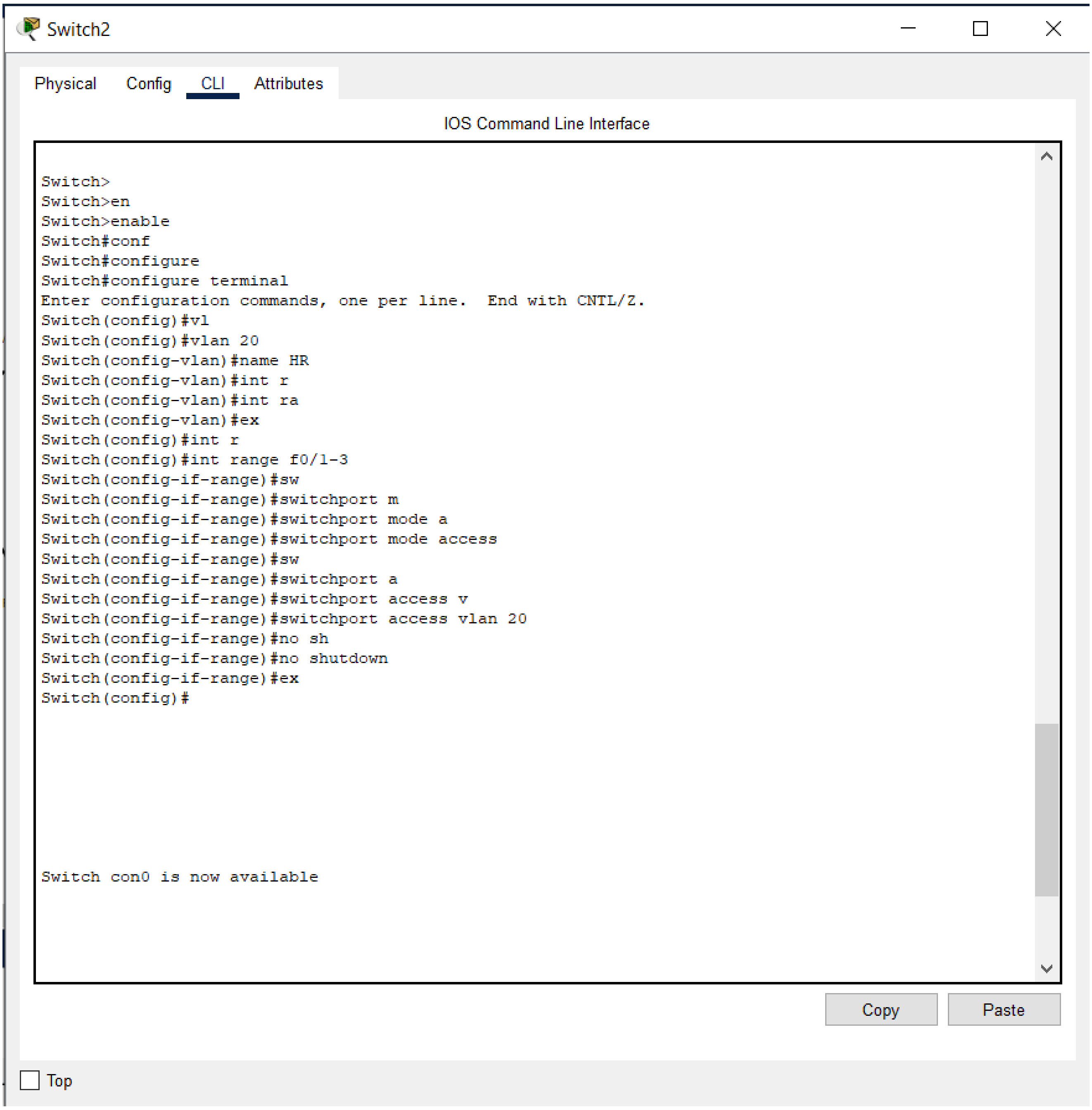
Variable Subnetting

Vlan_no	Network	Gateway/First	Last	Subnet mask	broadcast
10	192.168.5.0/26	192.168.5.1	192.168.5.62	255.255.255.192 (/26)	192.168.5.63
20	192.168.5.64/26	192.168.5.65	192.168.5.126	255.255.255.192 (/26)	192.168.5.127
30	192.168.5.128/28	192.168.5.129	192.168.5.142	255.255.255.240 (/28)	192.168.5.143
40	192.168.5.144/28	192.168.5.145	192.168.5.158	255.255.255.240 (/28)	192.168.5.159
50	192.168.5.160/28	192.168.5.161	192.168.5.174	255.255.255.240 (/28)	192.168.5.175

Implementation

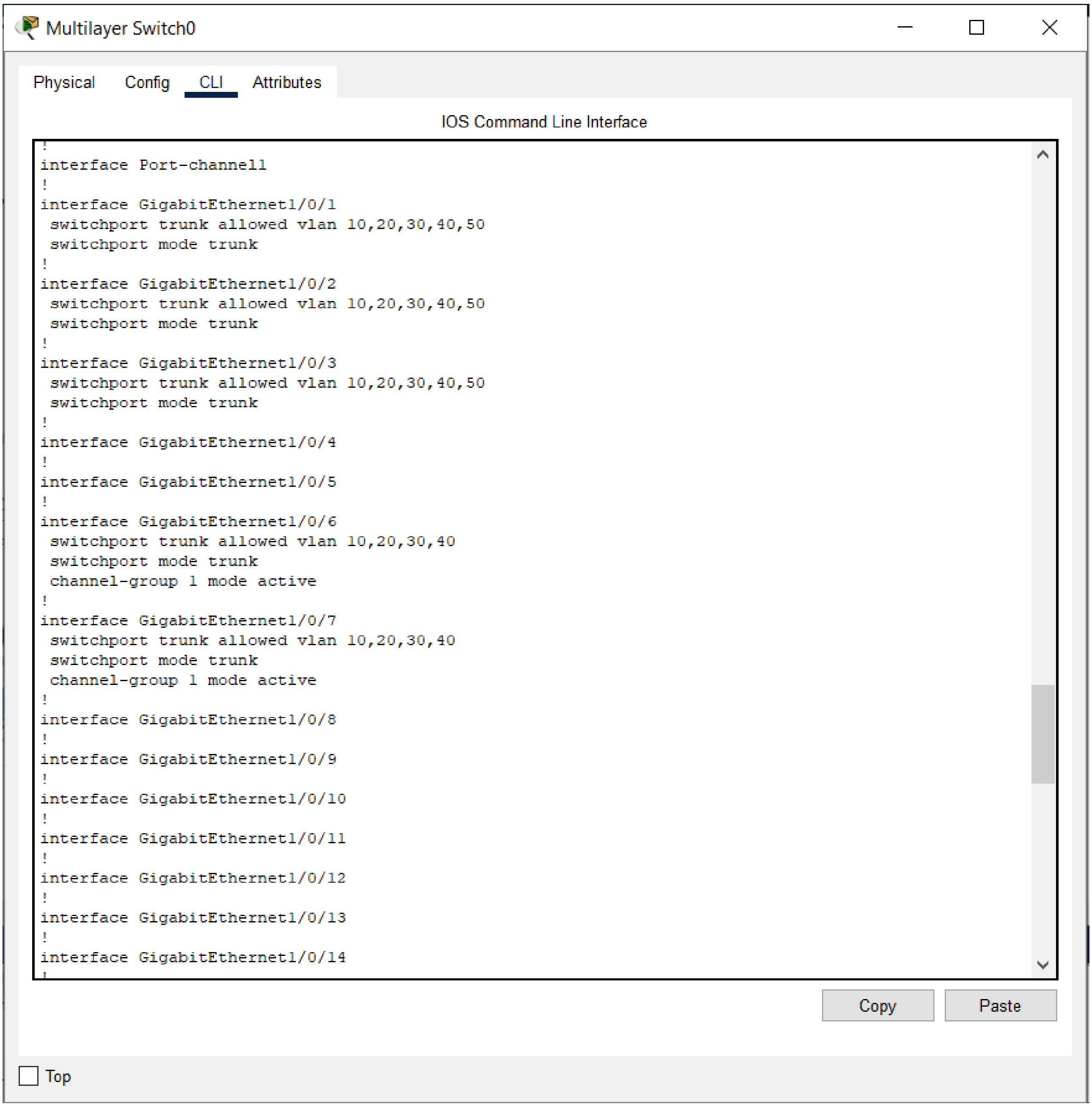
• Switchport mode





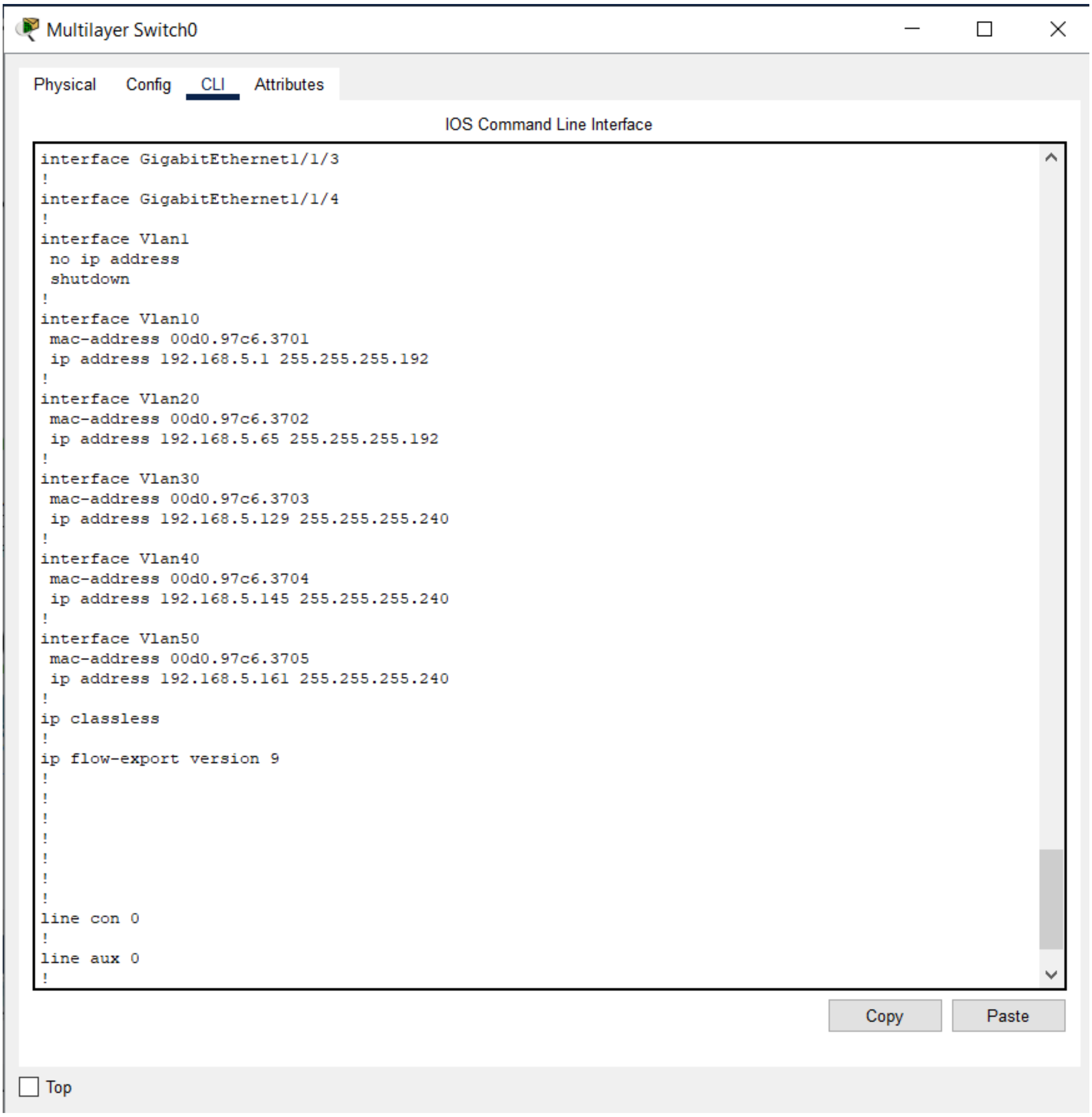
- ## EthreChannel

link aggregation technology that provides increased bandwidth capacity (Bandwidth Aggregation) and, crucially, redundancy. If one physical link in the bundle fails, the remaining links continue to pass data without network interruption


Multilayer Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
!
interface Port-channel1
!
interface GigabitEthernet1/0/1
switchport trunk allowed vlan 10,20,30,40,50
switchport mode trunk
!
interface GigabitEthernet1/0/2
switchport trunk allowed vlan 10,20,30,40,50
switchport mode trunk
!
interface GigabitEthernet1/0/3
switchport trunk allowed vlan 10,20,30,40,50
switchport mode trunk
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
channel-group 1 mode active
!
interface GigabitEthernet1/0/7
switchport trunk allowed vlan 10,20,30,40
switchport mode trunk
channel-group 1 mode active
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
!
interface GigabitEthernet1/0/11
!
interface GigabitEthernet1/0/12
!
interface GigabitEthernet1/0/13
!
interface GigabitEthernet1/0/14
Copy
Paste
Top

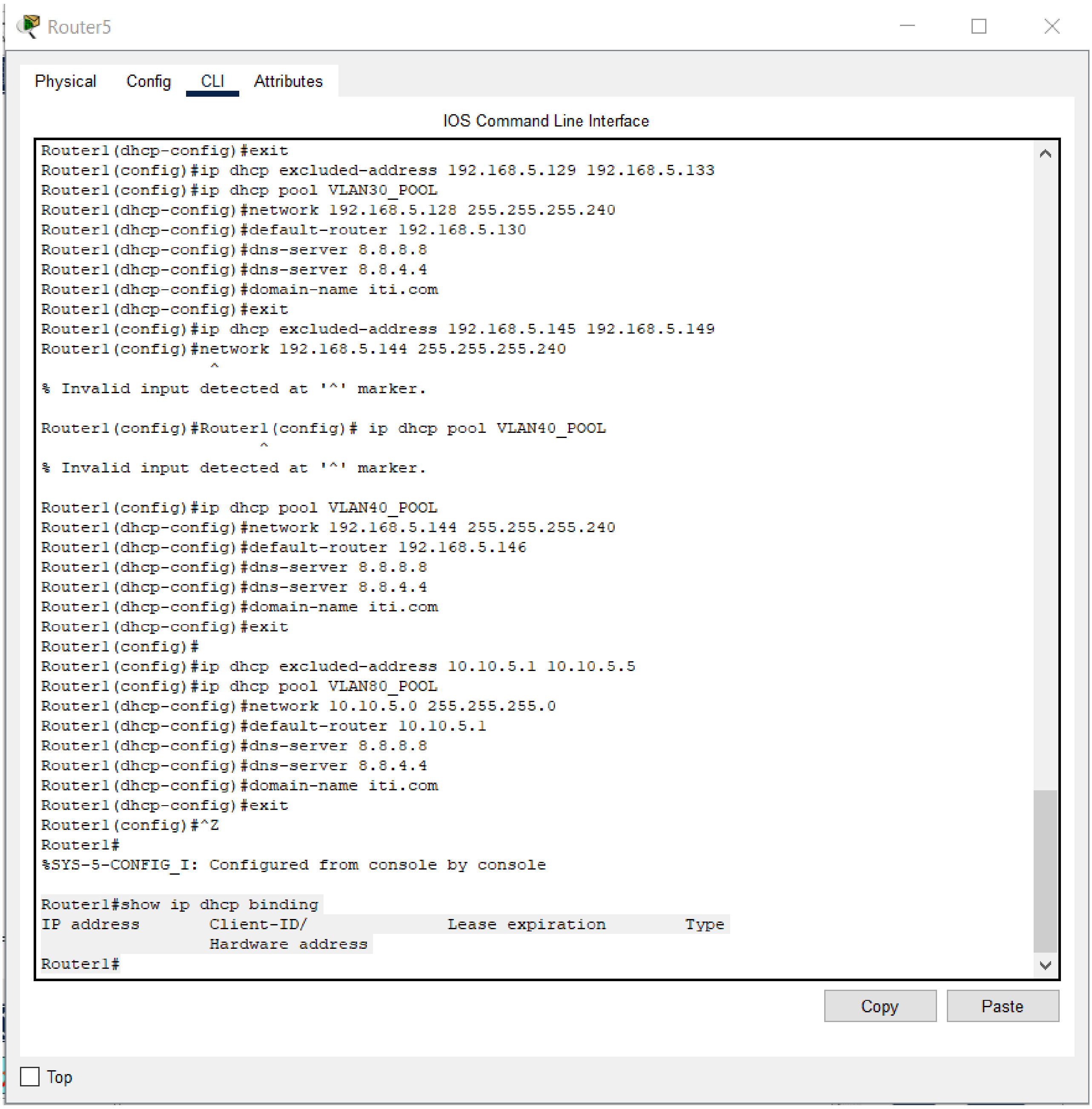
- ## VLANs

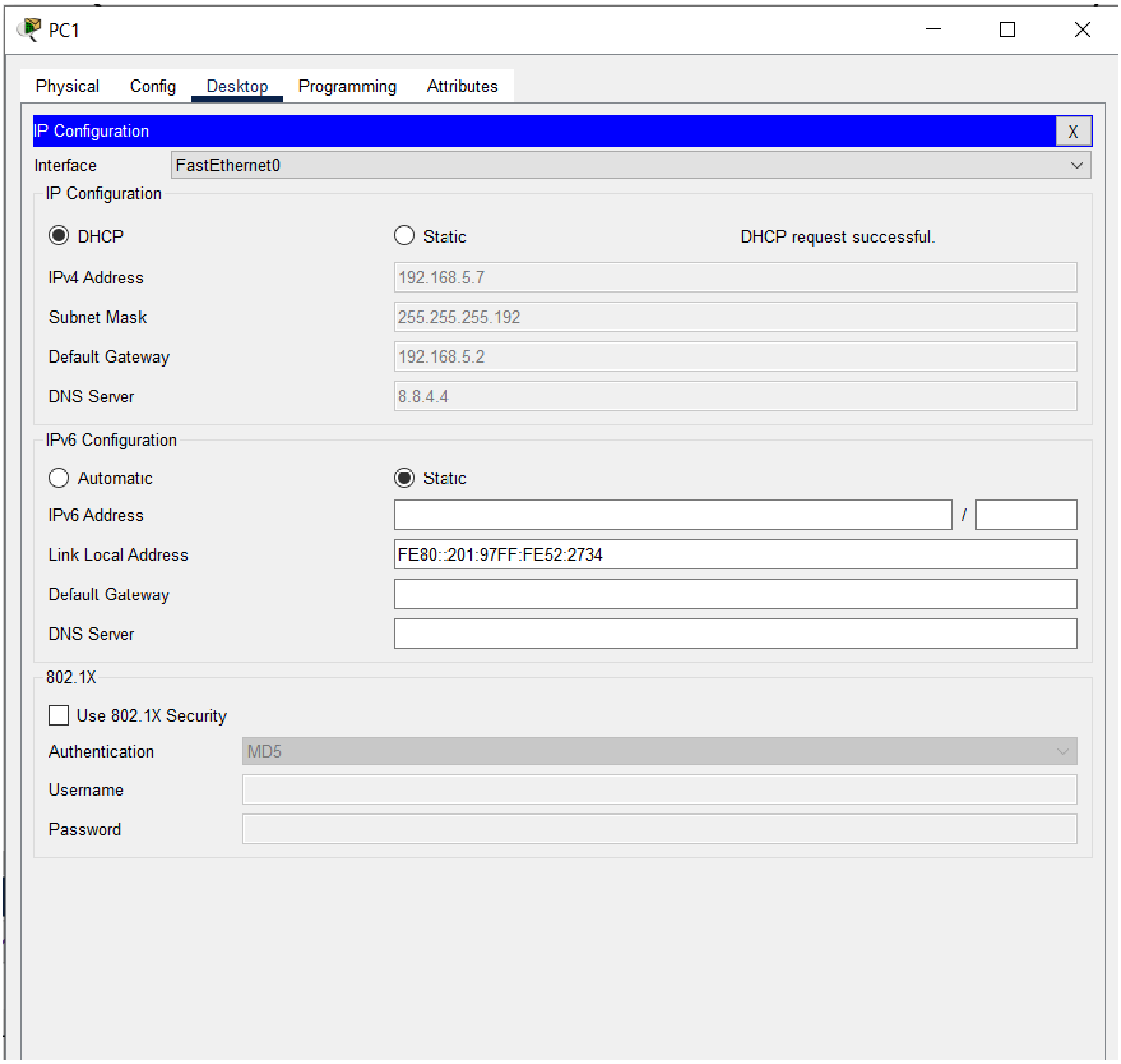
segment the Local Area Network (LAN) into smaller broadcast domains, isolating traffic for enhanced security and improved network performance



- ## DHCP

used to automatically distribute IP addresses, Default Gateway, and DNS server addresses to end devices (Hosts). Using a stand-alone server centralizes and simplifies IP address management




PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
DHCP request successful.
IPv4 Address
192.168.5.7
Subnet Mask
255.255.255.192
Default Gateway
192.168.5.2
DNS Server
8.8.4.4
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:97FF:FE52:2734
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password

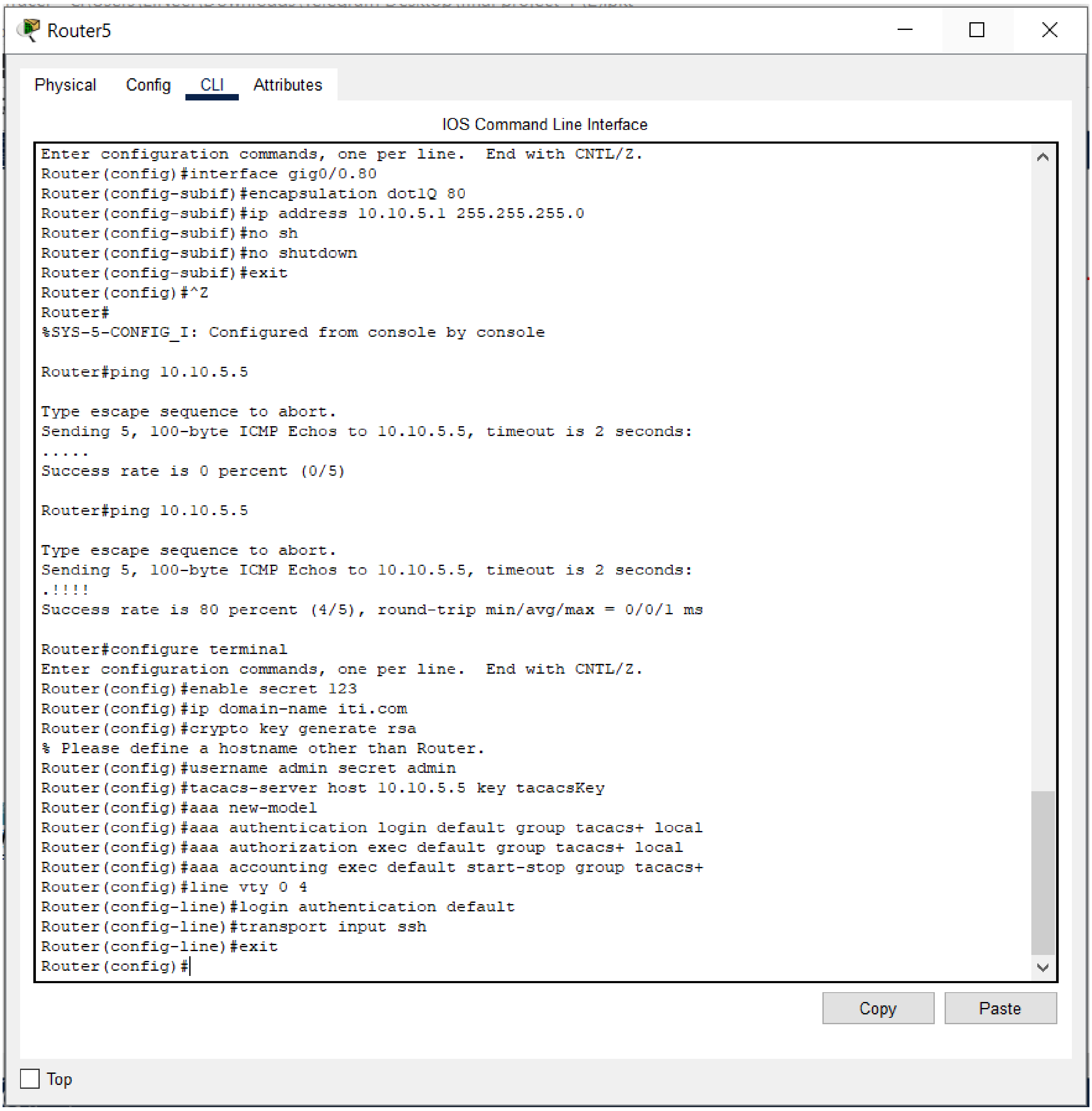
• AAA

Description: AAA (Authentication, Authorization, and Accounting) provides a secure and controlled method for accessing network devices

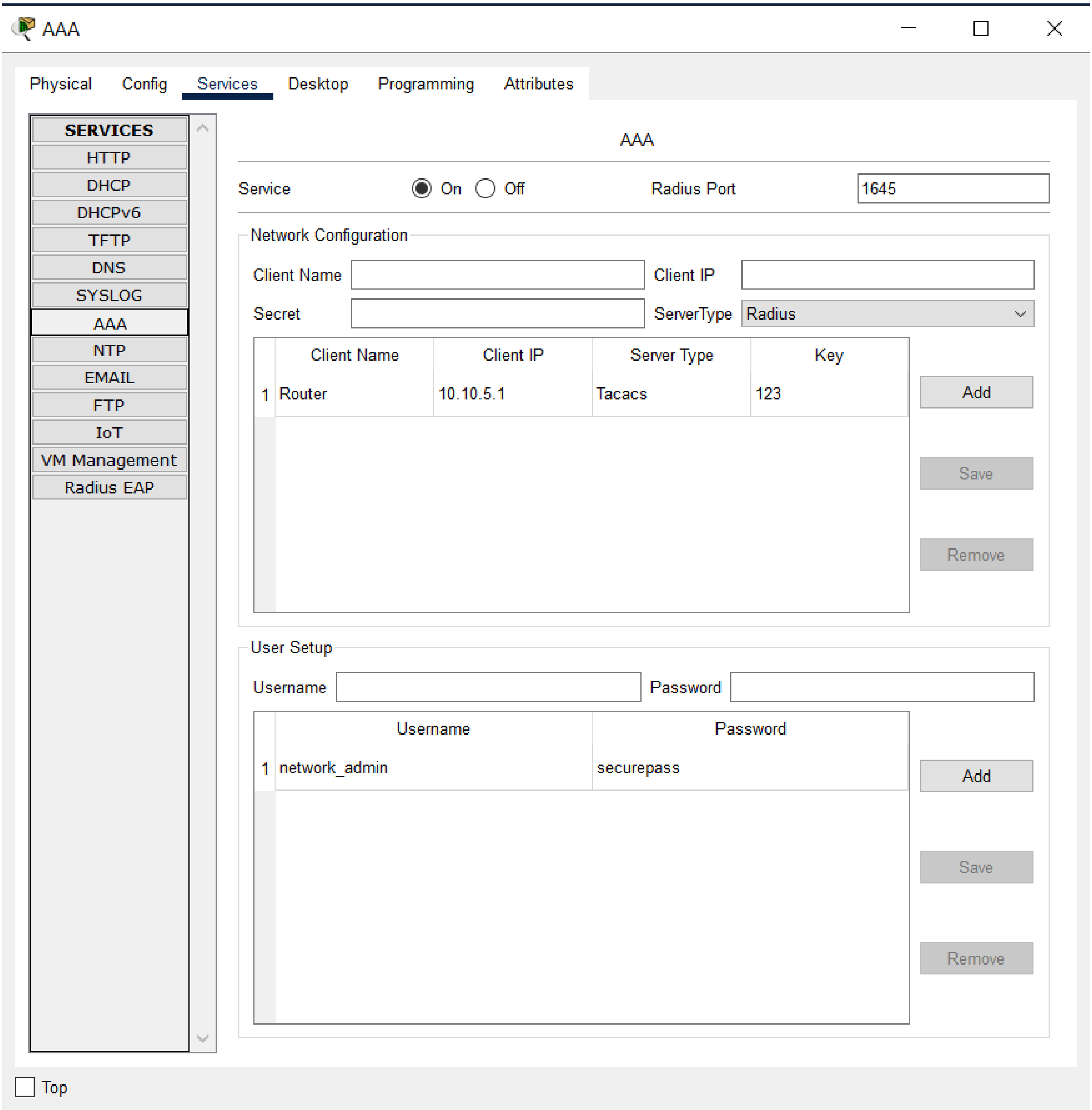
aaa new-model

aaa authentication login default group tacacs+ local

username admin secret admin



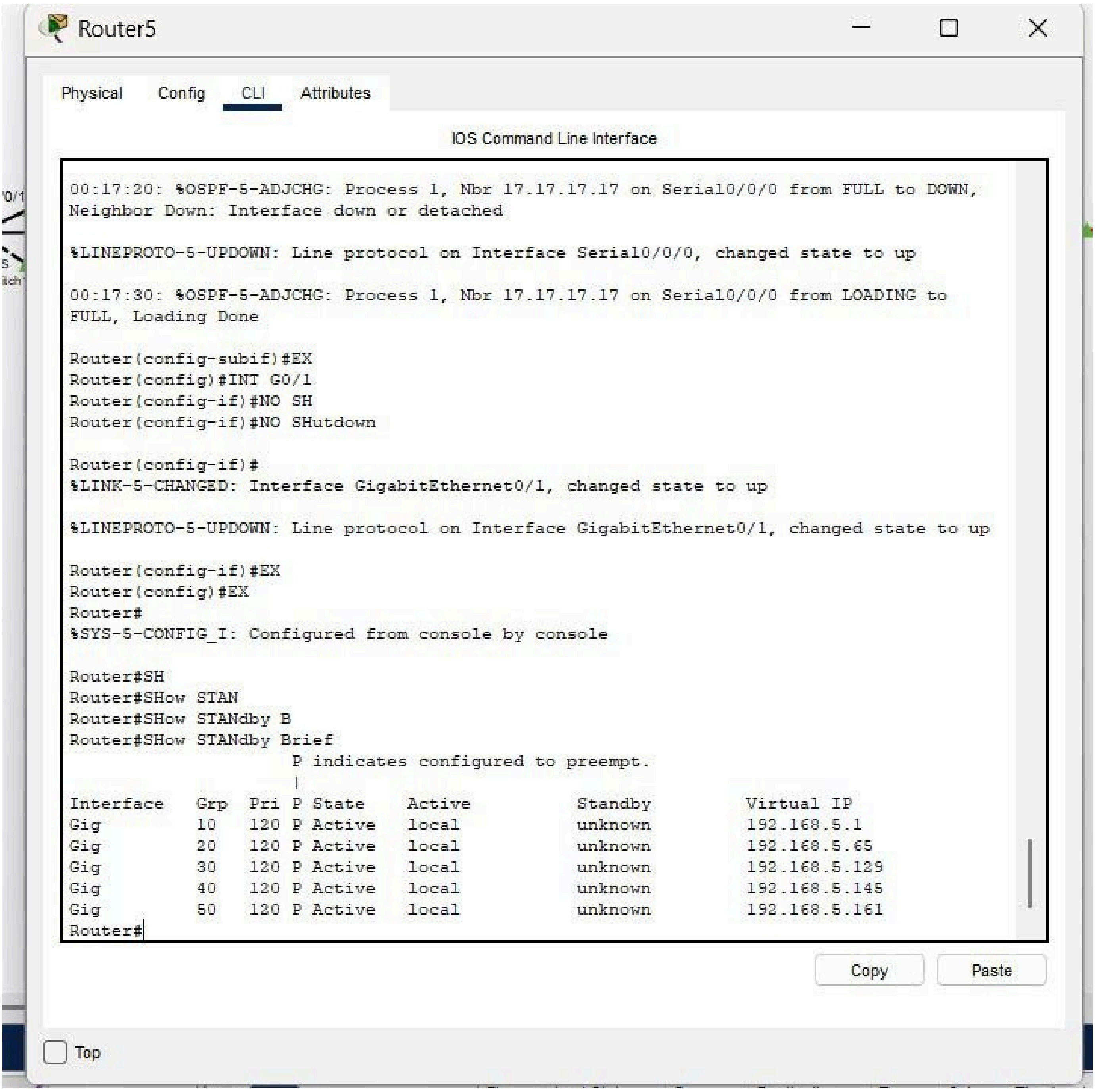

```
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0.80
Router(config-subif)#encapsulation dot1Q 80
Router(config-subif)#ip address 10.10.5.1 255.255.255.0
Router(config-subif)#no sh
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 10.10.5.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.5.5, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#ping 10.10.5.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.5.5, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#enable secret 123
Router(config)#ip domain-name iti.com
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
Router(config)#username admin secret admin
Router(config)#tacacs-server host 10.10.5.5 key tacacsKey
Router(config)#aaa new-model
Router(config)#aaa authentication login default group tacacs+ local
Router(config)#aaa authorization exec default group tacacs+ local
Router(config)#aaa accounting exec default start-stop group tacacs+
Router(config)#line vty 0 4
Router(config-line)#login authentication default
Router(config-line)#transport input ssh
Router(config-line)#exit
Router(config)#
```


AAA
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
AAA
Service
On
Off
Radius Port
1645
Network Configuration
Client Name
Client IP
Secret
ServerType
Radius
Client Name
Client IP
Server Type
Key
1
Router
10.10.5.1
Tacacs
123
Add
Save
Remove
User Setup
Username
Password
Username
Password
1
network_admin
securepass
Add
Save
Remove
Top

- # HSRP

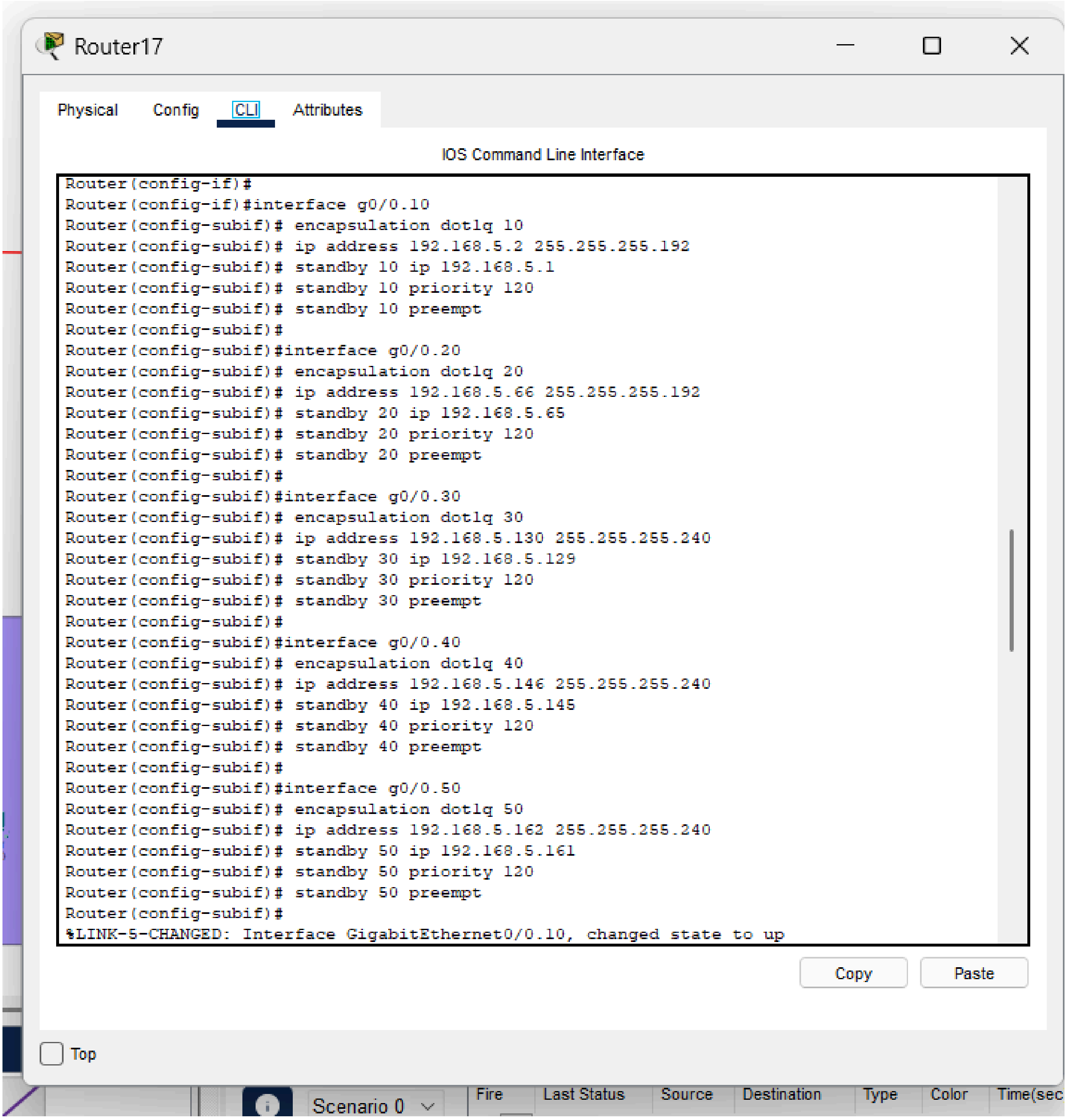
It is used to ensure high network availability by providing automatic failover if the primary router fails.

- A single Virtual IP Address is shared between two routers (R1 and R2).
- One router is designated as Active, and the other is Standby.
- If the Active router fails, the Standby router instantly takes over the role of the default gateway without requiring host configuration changes



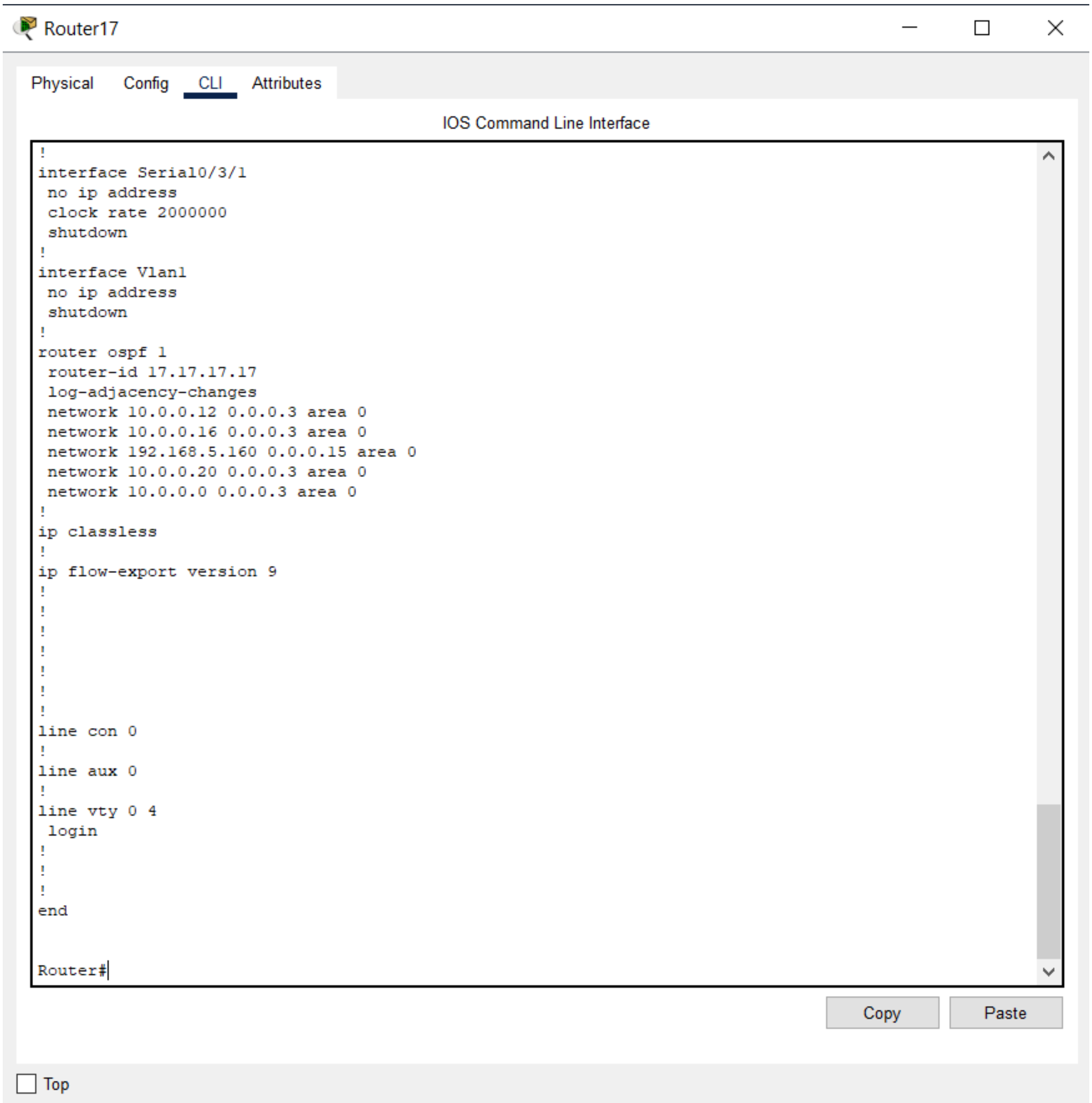
- ## HSRP

In an HSRP group, routers elect roles based on priority: one becomes Active, forwarding traffic, and the other becomes Standby, ready to take over instantly if needed. They share a common virtual MAC and virtual IP, so end devices always send traffic to the same logical gateway without knowing which physical router is handling it. HSRP uses periodic hello messages between routers to monitor availability. If the Standby router stops hearing the Active router’s hellos, it automatically becomes the new Active, providing a smooth transition with minimal downtime. Traffic forwarding resumes quickly without requiring client reconfiguration or manual intervention. This redundancy model protects the first-hop connectivity for LAN and VLAN networks, eliminating single points of failure at the gateway level. It works alongside dynamic routing and link aggregation to strengthen both path redundancy and device reliability.


Router17
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config-if)#
Router(config-if)#interface g0/0.10
Router(config-subif)# encapsulation dot1q 10
Router(config-subif)# ip address 192.168.5.2 255.255.255.192
Router(config-subif)# standby 10 ip 192.168.5.1
Router(config-subif)# standby 10 priority 120
Router(config-subif)# standby 10 preempt
Router(config-subif)#
Router(config-subif)#interface g0/0.20
Router(config-subif)# encapsulation dot1q 20
Router(config-subif)# ip address 192.168.5.66 255.255.255.192
Router(config-subif)# standby 20 ip 192.168.5.65
Router(config-subif)# standby 20 priority 120
Router(config-subif)# standby 20 preempt
Router(config-subif)#
Router(config-subif)#interface g0/0.30
Router(config-subif)# encapsulation dot1q 30
Router(config-subif)# ip address 192.168.5.130 255.255.255.240
Router(config-subif)# standby 30 ip 192.168.5.129
Router(config-subif)# standby 30 priority 120
Router(config-subif)# standby 30 preempt
Router(config-subif)#
Router(config-subif)#interface g0/0.40
Router(config-subif)# encapsulation dot1q 40
Router(config-subif)# ip address 192.168.5.146 255.255.255.240
Router(config-subif)# standby 40 ip 192.168.5.145
Router(config-subif)# standby 40 priority 120
Router(config-subif)# standby 40 preempt
Router(config-subif)#
Router(config-subif)#interface g0/0.50
Router(config-subif)# encapsulation dot1q 50
Router(config-subif)# ip address 192.168.5.162 255.255.255.240
Router(config-subif)# standby 50 ip 192.168.5.161
Router(config-subif)# standby 50 priority 120
Router(config-subif)# standby 50 preempt
Router(config-subif)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0.10, changed state to up
Copy
Paste
Top
Scenario 0
Fire
Last Status
Source
Destination
Type
Color
Time(sec

- ## OSPF

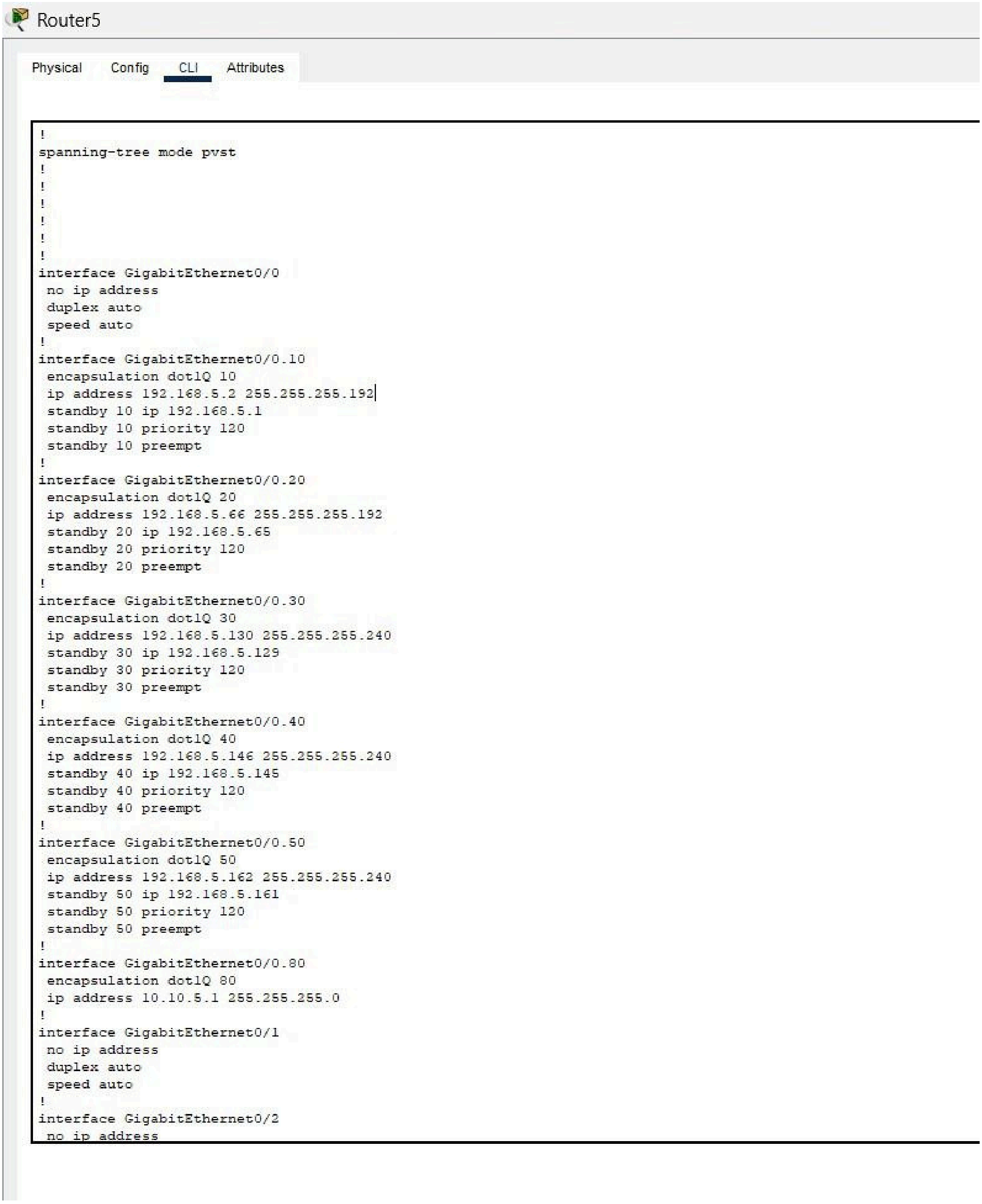
I configured OSPF (Open Shortest Path First) as an Interior Gateway Protocol (IGP) that relies on the Link-State algorithm to determine the fastest paths based on calculated cost. We use OSPF to ensure that all routers within our internal network have comprehensive knowledge of all subnets (VLANs and Point-to-Point links) and can route traffic efficiently and quickly




```
!
interface Serial0/3/1
 no ip address
 clock rate 2000000
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router ospf 1
 router-id 17.17.17.17
 log-adjacency-changes
 network 10.0.0.12 0.0.0.3 area 0
 network 10.0.0.16 0.0.0.3 area 0
 network 192.168.5.160 0.0.0.15 area 0
 network 10.0.0.20 0.0.0.3 area 0
 network 10.0.0.0 0.0.0.3 area 0
!
ip classless
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end

Router#
```

- ## ROAS

I configured sub-interfaces on the router to enable inter-VLAN routing using the "Router on a Stick" method, allowing devices in different VLANs to communicate securely.It utilizes a single physical interface on the router (e.g., Gi0/0), which is logically divided into subinterfaces. Each subinterface is assigned to a specific VLAN using the **encapsulation dot1Q** protocol for tagging

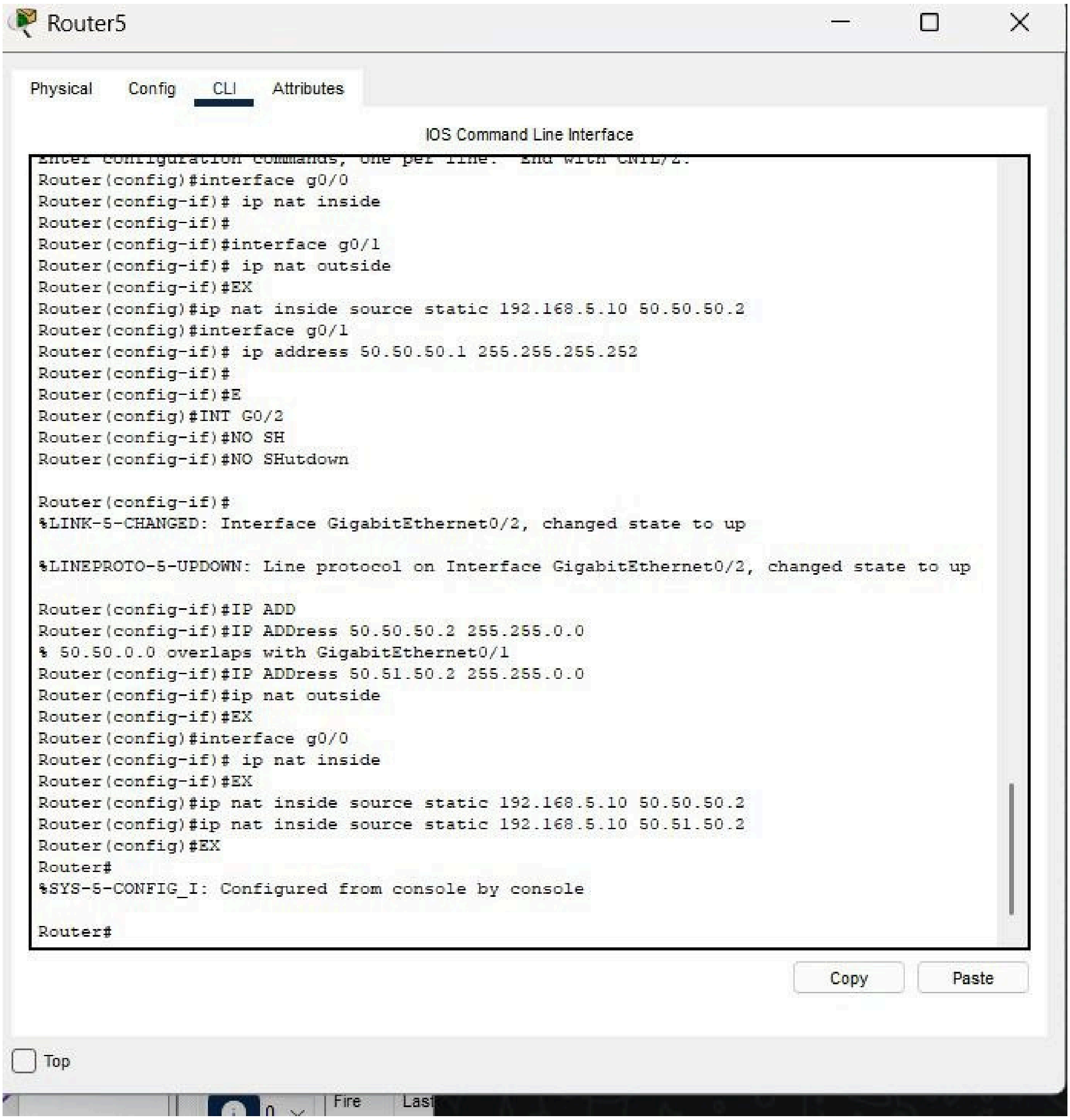


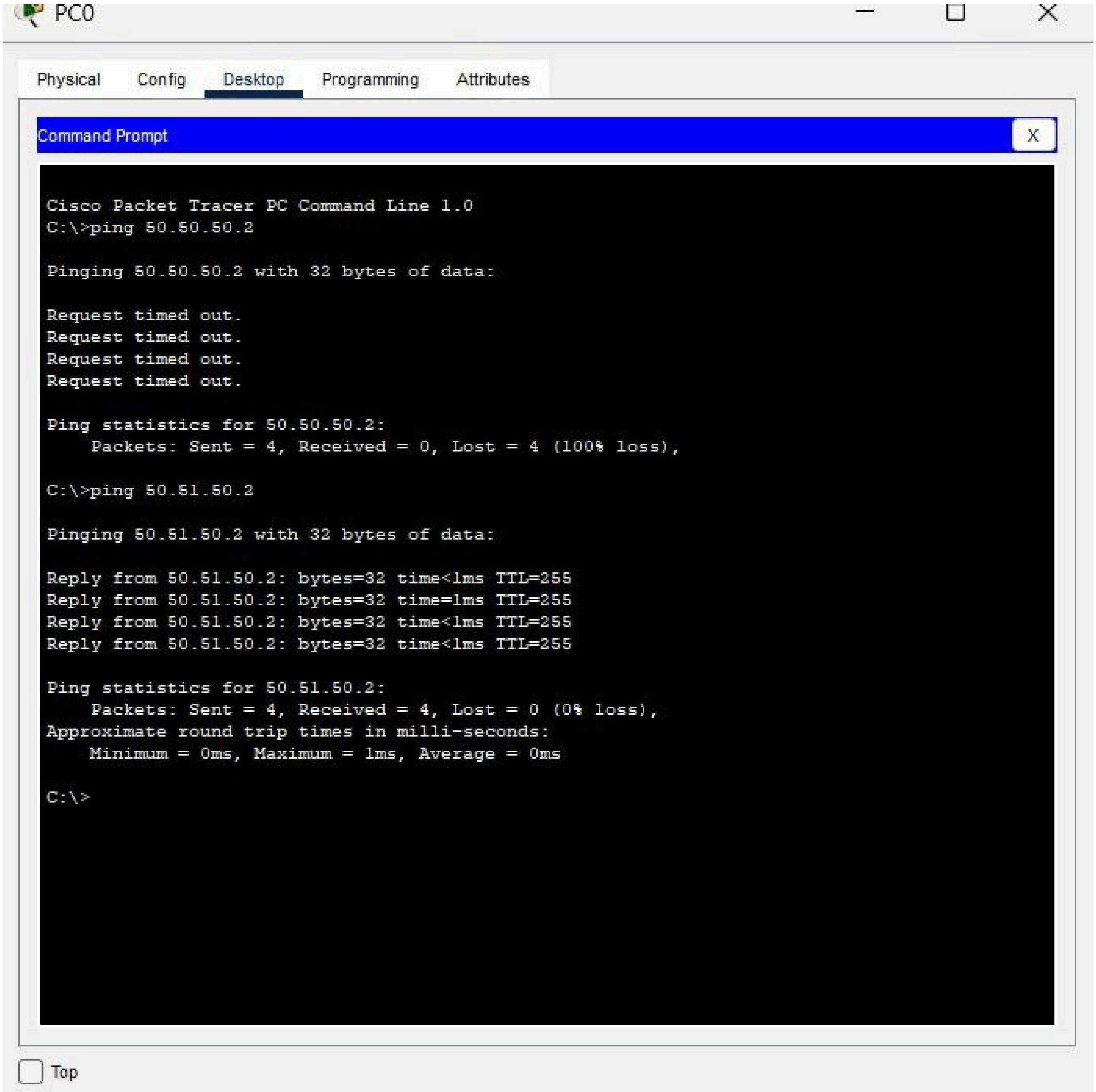

```
!
spanning-tree mode pvst
!
!
!
!
!
!
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.5.2 255.255.255.192
 standby 10 ip 192.168.5.1
 standby 10 priority 120
 standby 10 preempt
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.5.66 255.255.255.192
 standby 20 ip 192.168.5.65
 standby 20 priority 120
 standby 20 preempt
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.5.130 255.255.255.240
 standby 30 ip 192.168.5.129
 standby 30 priority 120
 standby 30 preempt
!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.5.146 255.255.255.240
 standby 40 ip 192.168.5.145
 standby 40 priority 120
 standby 40 preempt
!
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 192.168.5.162 255.255.255.240
 standby 50 ip 192.168.5.161
 standby 50 priority 120
 standby 50 preempt
!
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 10.10.5.1 255.255.255.0
!
interface GigabitEthernet0/1
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 no ip address
```

- ## Nat

used to create a one-to-one, persistent mapping between an internal private IP address and a static external public IP address.

In this project, Static NAT is configured on Router R2 to allow external users (Internet) to consistently access our public Web Server. Any incoming request directed to the public IP address will be translated to the server's private IP address.




PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 50.50.50.2

Pinging 50.50.50.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 50.50.50.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 50.51.50.2

Pinging 50.51.50.2 with 32 bytes of data:

Reply from 50.51.50.2: bytes=32 time<1ms TTL=255
Reply from 50.51.50.2: bytes=32 time=1ms TTL=255
Reply from 50.51.50.2: bytes=32 time<1ms TTL=255
Reply from 50.51.50.2: bytes=32 time<1ms TTL=255

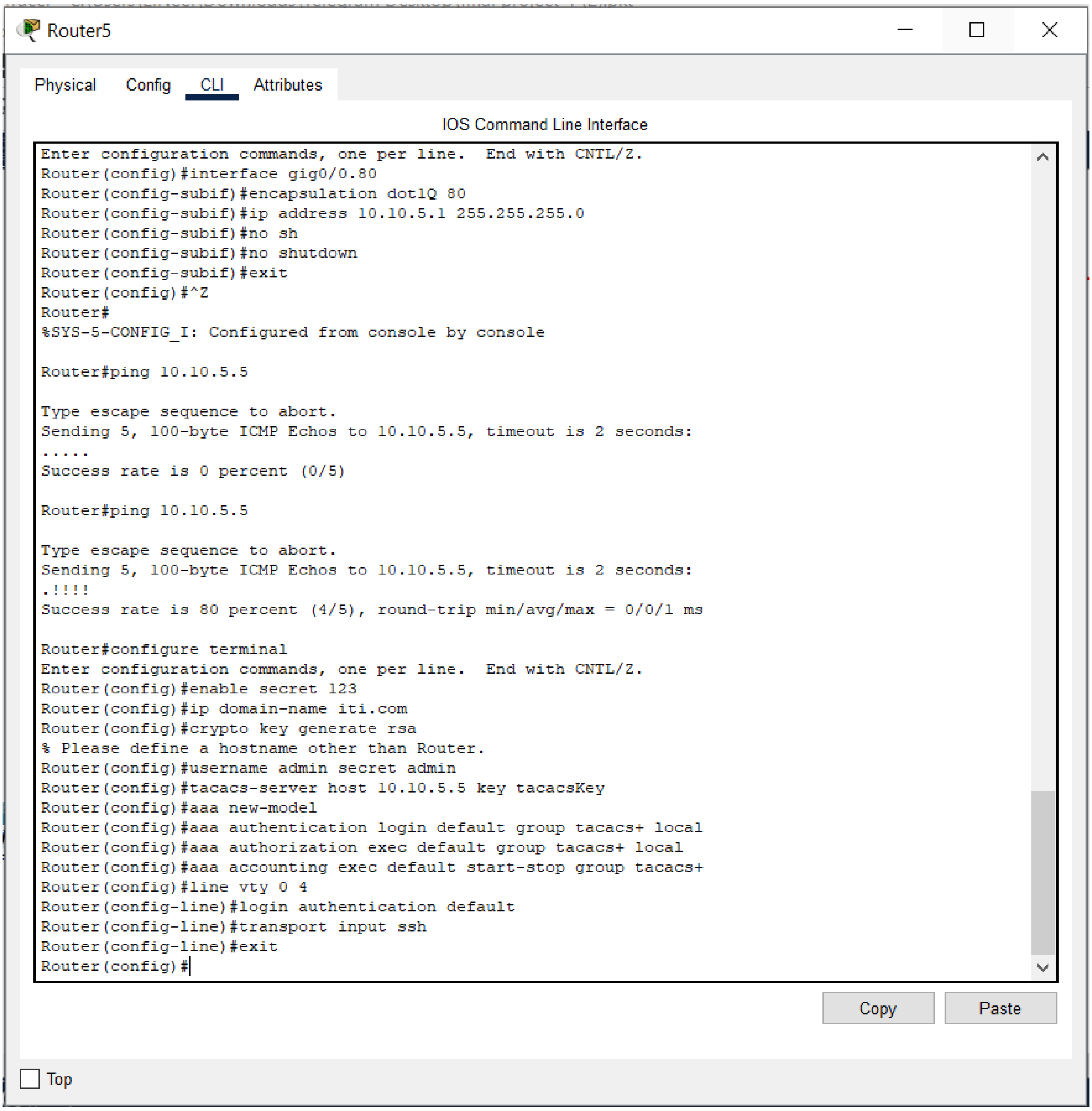
Ping statistics for 50.51.50.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
Top

- ## SSH

SSH provide secure device management.

```
ip domain-name iti.com
username admin secret password123
crypto key generate rsa
How many bits in the modulus [512]: 2048
line vty 0 4
transport input ssh
login local
exit
enable secret securepass
```




```
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/0.80
Router(config-subif)#encapsulation dot1Q 80
Router(config-subif)#ip address 10.10.5.1 255.255.255.0
Router(config-subif)#no sh
Router(config-subif)#no shutdown
Router(config-subif)#exit
Router(config)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 10.10.5.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.5.5, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#ping 10.10.5.5

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.5.5, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#enable secret 123
Router(config)#ip domain-name iti.com
Router(config)#crypto key generate rsa
% Please define a hostname other than Router.
Router(config)#username admin secret admin
Router(config)#tacacs-server host 10.10.5.5 key tacacsKey
Router(config)#aaa new-model
Router(config)#aaa authentication login default group tacacs+ local
Router(config)#aaa authorization exec default group tacacs+ local
Router(config)#aaa accounting exec default start-stop group tacacs+
Router(config)#line vty 0 4
Router(config-line)#login authentication default
Router(config-line)#transport input ssh
Router(config-line)#exit
Router(config)#
```